

FALL 2026



# EE3003 Analog and Digital Communications

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Department of Electrical Engineering

FALL 2026

# Course Learning Outcomes

| No. | Course Learning Outcome (CLO) Statements                                                                                                                                                      | Assessment Tools |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1   | <b>Explain</b> the fundamental models of communication systems, signal transmission characteristics, distortions, and energy/power spectral densities.                                        | Q1, A1, M1       |
| 2   | <b>Calculate</b> parameters like bandwidth, power efficiency, transmission efficiency, and modulation index for linear (AM, SSB, VSB, QAM) and non-linear (FM, PM) analog modulation systems. | Q2, M1, M2       |
| 3   | <b>Apply</b> the sampling theorem and analog-to-digital pulse conversion techniques (PAM, PWM, PPM, PCM, DPCM, DM) for baseband processing.                                                   | A2, M2, F        |
| 4   | <b>Analyze</b> the performance, error probabilities, and spectral characteristics of digital communication systems while evaluating fundamental information theory trade-offs.                | Q3, A3(PBL), F   |

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# Lecture Objectives

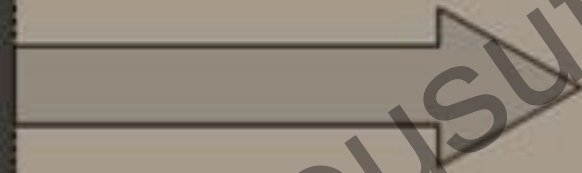
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- Signal Transmission through an LTI
- Distortionless Transmission
- Ideal and Practical Filters
- Signal Distortion Over a Communication Channel
  - Linear Distortion
  - Spectral Dispersion

Input



Linear  
System



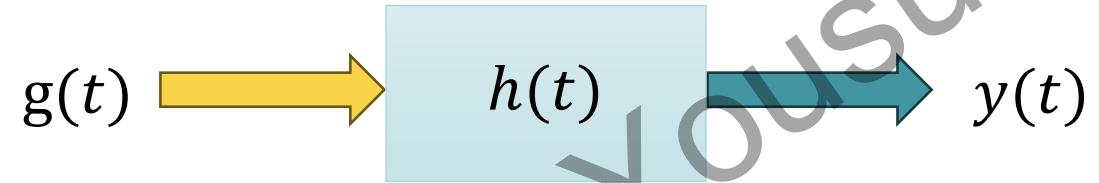
Output

# Linear Time Invariant (LTI) System

| Input                         | Output                        |
|-------------------------------|-------------------------------|
| $x_i(t)$                      | $y_i(t)$                      |
| $a_i \cdot x_i(t)$            | $a_i \cdot y_i(t)$            |
| $a \cdot x(t) + a \cdot x(t)$ | $a \cdot y(t) + a \cdot y(t)$ |

# Linear Time Invariant (LTI) System

## Time Domain Representation



- For a linear, time-invariant, continuous-time system; the input-output relationship is given by

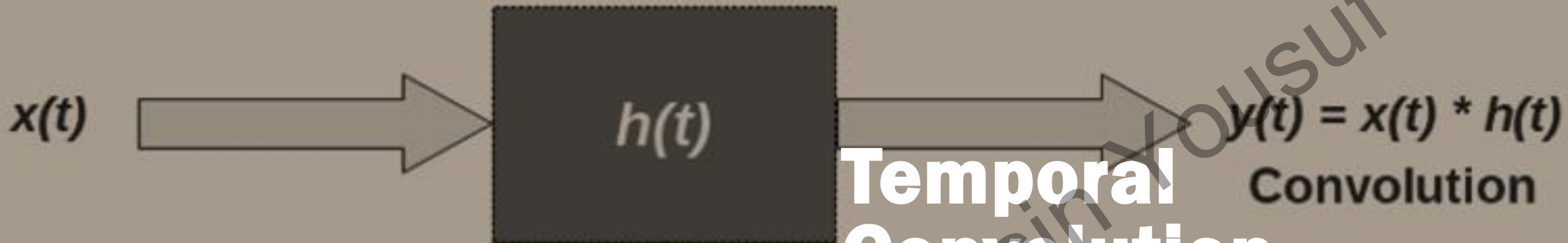
$$\int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

$$y(t) = g(t) * h(t) = \int_{-\infty}^{\infty} g(\tau) h(t - \tau) d\tau$$

- If  $g(t) = \delta(t)$ , i.e. unit impulse, then  $y(t) = \delta(t) * h(t)$  corresponds to;  $y(t) = h(t)$ .
- $h(t)$  is the **unit impulse response of the LTI system**.
- If  $g(t) \Leftrightarrow G(\omega)$ ,  $y(t) \Leftrightarrow Y(\omega)$ , and  $h(t) \Leftrightarrow H(\omega)$   
$$Y(\omega) = G(\omega)H(\omega)$$

where  $H(\omega)$  is the **system transfer function**.

**Convolution in time-domain corresponds to multiplication in frequency-domain**

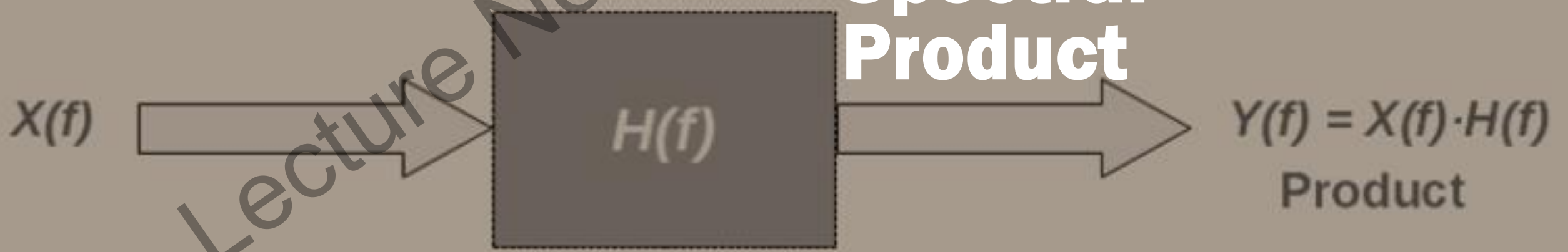


# Temporal Convolution

FOURIER  
TRANSFORM



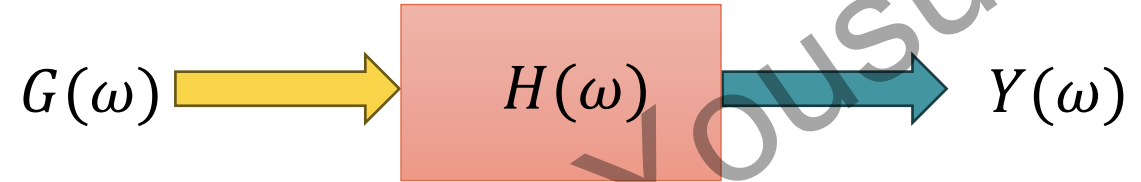
=  
INVERSE  
FOURIER  
TRANSFORM



# Spectral Product

# Linear Time Invariant (LTI) System

## Frequency Domain Representation



- The system transfer function,  $H(\omega)$  is the spectral response of the system as well.

$$Y(\omega) = G(\omega)H(\omega)$$

➤ In polar form,

$$|Y(\omega)|e^{j\theta_y(\omega)} = |G(\omega)|e^{j\theta_g(\omega)}|H(\omega)|e^{j\theta_h(\omega)}, \text{ and}$$

$$|Y(\omega)|e^{j\theta_y(\omega)} = |G(\omega)||H(\omega)|e^{j[\theta_g(\omega)+\theta_h(\omega)]}$$

➤ Therefore,

$$|Y(\omega)| = |G(\omega)||H(\omega)|$$

$$\theta_y(\omega) = \theta_g(\omega) + \theta_h(\omega)$$

Amplitude Response

Phase Response

The plots of  $|H(\omega)|$  and  $\theta_h(\omega)$  as a function of  $\omega$  is known as frequency response.

# Distortionless Transmission through an LTI Channel

$$|H(\omega)| = k, \text{ Amplitude Response}$$

$$\theta_h(\omega) = -\omega t_d, \text{ Phase Response}$$

Transmission is said to be **distortionless** if the input and output have **identical wave shapes** within a **multiplicative constant**.

- For e.g., a delayed output that retains the input waveform

$$y(t) = kg(t - t_d)$$

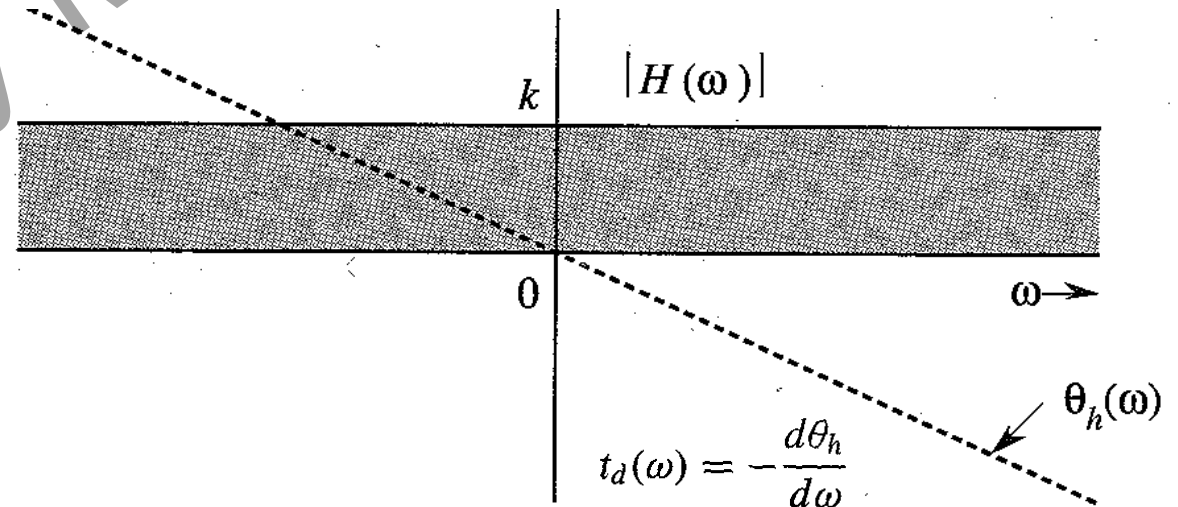
The Fourier Transform is,

$$Y(\omega) = kG(\omega)e^{-j\omega t_d}$$

But,  $Y(\omega) = G(\omega)H(\omega)$

Therefore, the transfer function,  $H(\omega)$  for distortionless transmission is,  $H(\omega) = k e^{-j\omega t_d}$

A system may have a flat amplitude response and yet distort a signal beyond recognition if the phase response is not linear i.e.  $t_d$  not constant.



**Figure 3.26** Linear time-invariant system frequency response for distortionless transmission.

**EXAMPLE 3.16** If  $g(t)$  and  $y(t)$  are the input and the output, respectively, of a simple RC low-pass filter (Fig. 3.27a), determine the transfer function  $H(\omega)$  and sketch  $|H(\omega)|$ ,  $\theta_h(\omega)$ , and  $t_d(\omega)$ . For distortionless transmission through this filter, what is the requirement on the bandwidth of  $g(t)$  if amplitude response variation within 2% and time delay variation within 5% are tolerable? What is the transmission delay? Find the output  $y(t)$ .

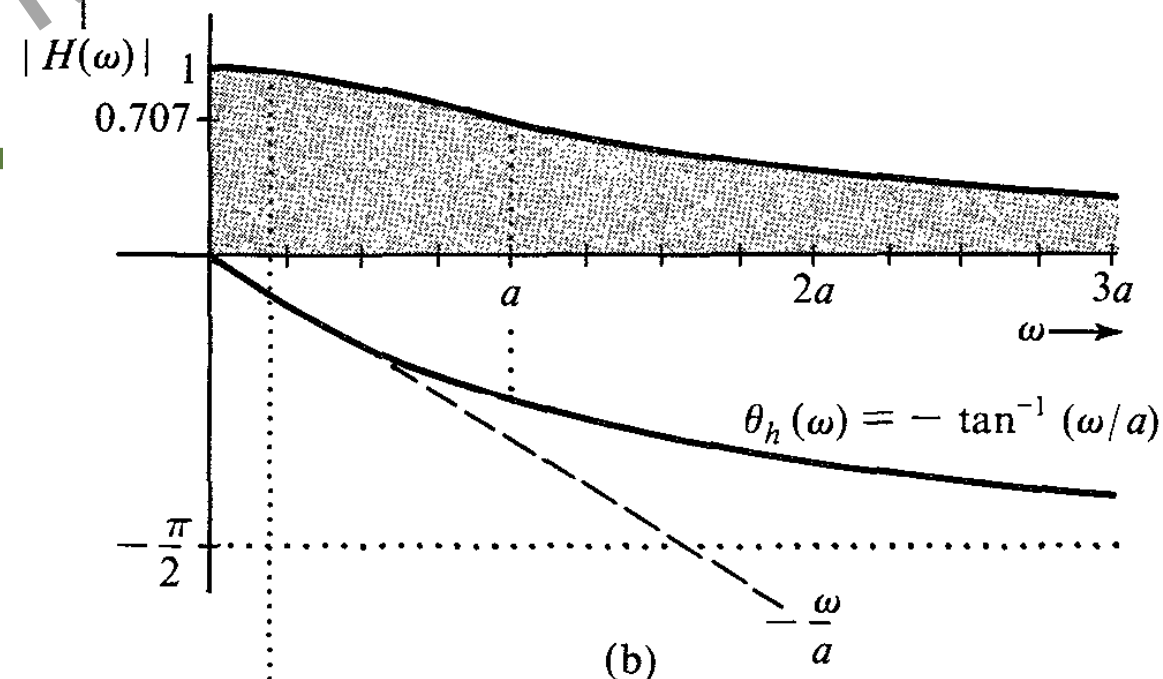
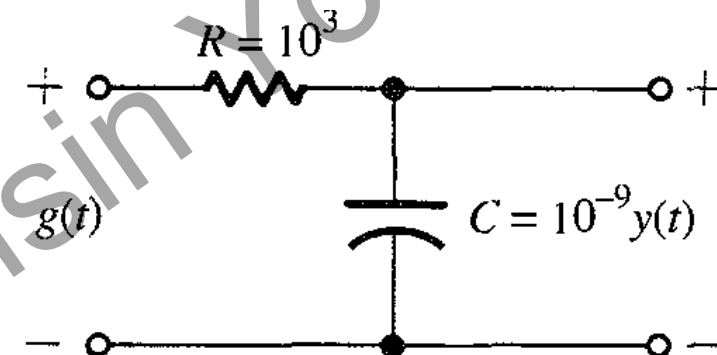
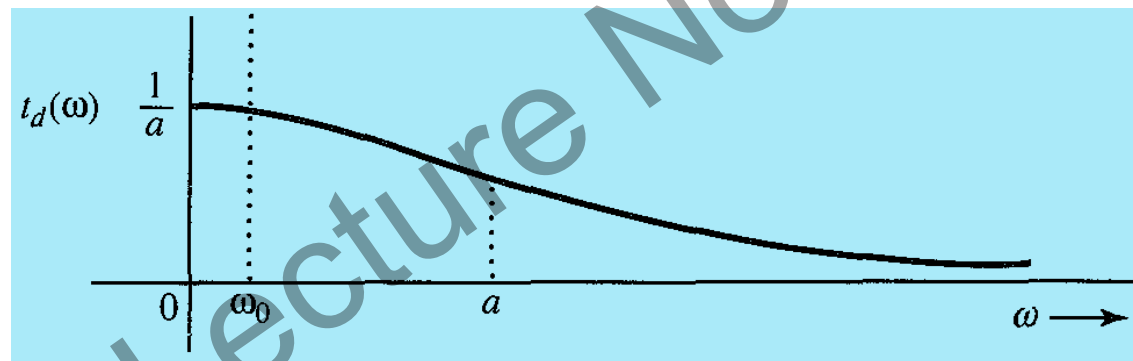
Application of the voltage division rule to this circuit yields

$$H(\omega) = \frac{1/j\omega C}{R + (1/j\omega C)} = \frac{1}{1 + j\omega RC} = \frac{a}{a + j\omega}$$

$$|H(\omega)| = \frac{a}{\sqrt{a^2 + \omega^2}} \simeq 1 \quad \omega \ll a \quad a = \frac{1}{RC} = 10^6$$

$$\theta_h(\omega) = -\tan^{-1} \frac{\omega}{a} \simeq -\frac{\omega}{a} \quad \omega \ll a$$

$$t_d(\omega) = -\frac{d\theta_h}{d\omega} = \frac{a}{\omega^2 + a^2} \simeq \frac{1}{a} = 10^{-6} \quad \omega \ll a$$



The filter can thus transmit low-frequency signals with negligible distortion.

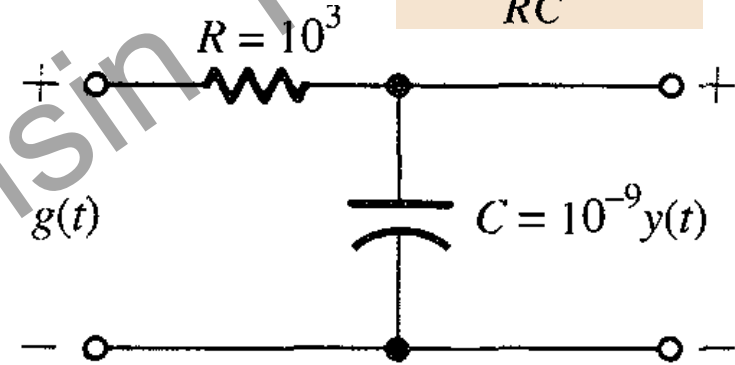
**EXAMPLE 3.16** If  $g(t)$  and  $y(t)$  are the input and the output, respectively, of a simple  $RC$  low-pass filter (Fig. 3.27a), determine the transfer function  $H(\omega)$  and sketch  $|H(\omega)|$ ,  $\theta_h(\omega)$ , and  $t_d(\omega)$ . For distortionless transmission through this filter, what is the requirement on the bandwidth of  $g(t)$  if amplitude response variation within 2% and time delay variation within 5% are tolerable? What is the transmission delay? Find the output  $y(t)$ .

$$|H(\omega)| = \frac{a}{\sqrt{a^2 + \omega^2}}$$

$$\theta_h(\omega) = -\tan^{-1} \frac{\omega}{a}$$

$$t_d(\omega) = -\frac{d\theta_h}{d\omega} = \frac{a}{\omega^2 + a^2}$$

$$a = \frac{1}{RC} = 10^6$$



- Let  $\omega_0$  be the highest bandwidth of a signal that can be transmitted within 2% of amplitude variation and time delay variation within 5%.
- For a LPF, the maximum values for gain and time delay will be at  $\omega = 0$ .
- $|H(0)| = 1$  and  $t_d(0) = \frac{1}{a}$
- Therefore,  $|H(\omega_0)| \geq 0.98$  and  $t_d(\omega_0) \geq \frac{0.95}{a}$ , so that

$$|H(\omega_0)| = \frac{a}{\sqrt{\omega_0^2 + a^2}} \geq 0.98, \quad (i)$$

$$\omega_0 \leq 0.203a = 203,000$$

$$t_d(\omega_0) = \frac{a}{\omega_0^2 + a^2} \geq \frac{0.95}{a}, \quad (ii)$$

$$\omega_0 \leq 0.2294a = 229,400$$

The time-delay  $t_d \approx \frac{1}{a} = 10^{-6}$  over this band, and the amplitude response is almost unity, so the output is,

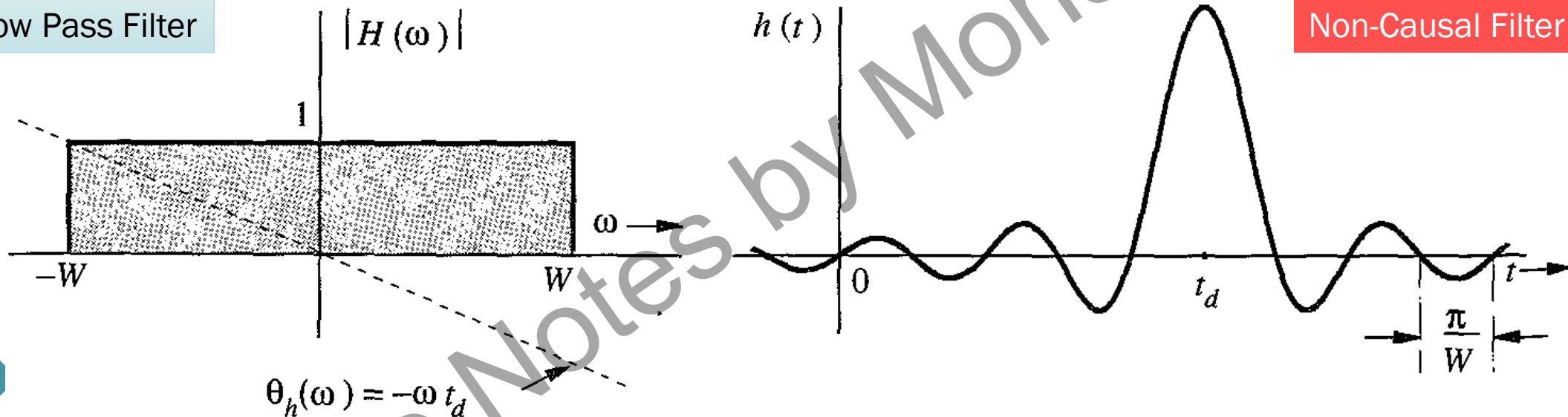
$$y(t) \approx g(t - t_d) = g(t - 10^{-6})$$

The smallest of the two values,  $\omega_0 = 203,000 \text{ rad/s}$  or  $32.31 \text{ kHz}$  is the highest bandwidth that satisfies both constraints on  $|H(\omega)|$  and  $t_d$ .

# Ideal and Practical Filters

Ideal filters **allow** distortionless transmission of a certain band of frequencies and **suppress** all the remaining frequencies as shown.

Low Pass Filter



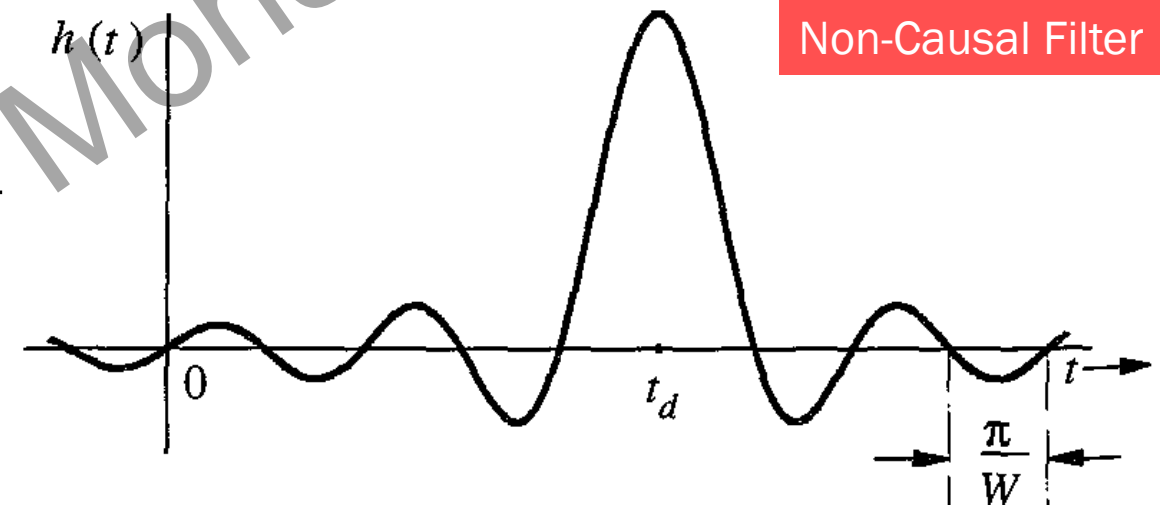
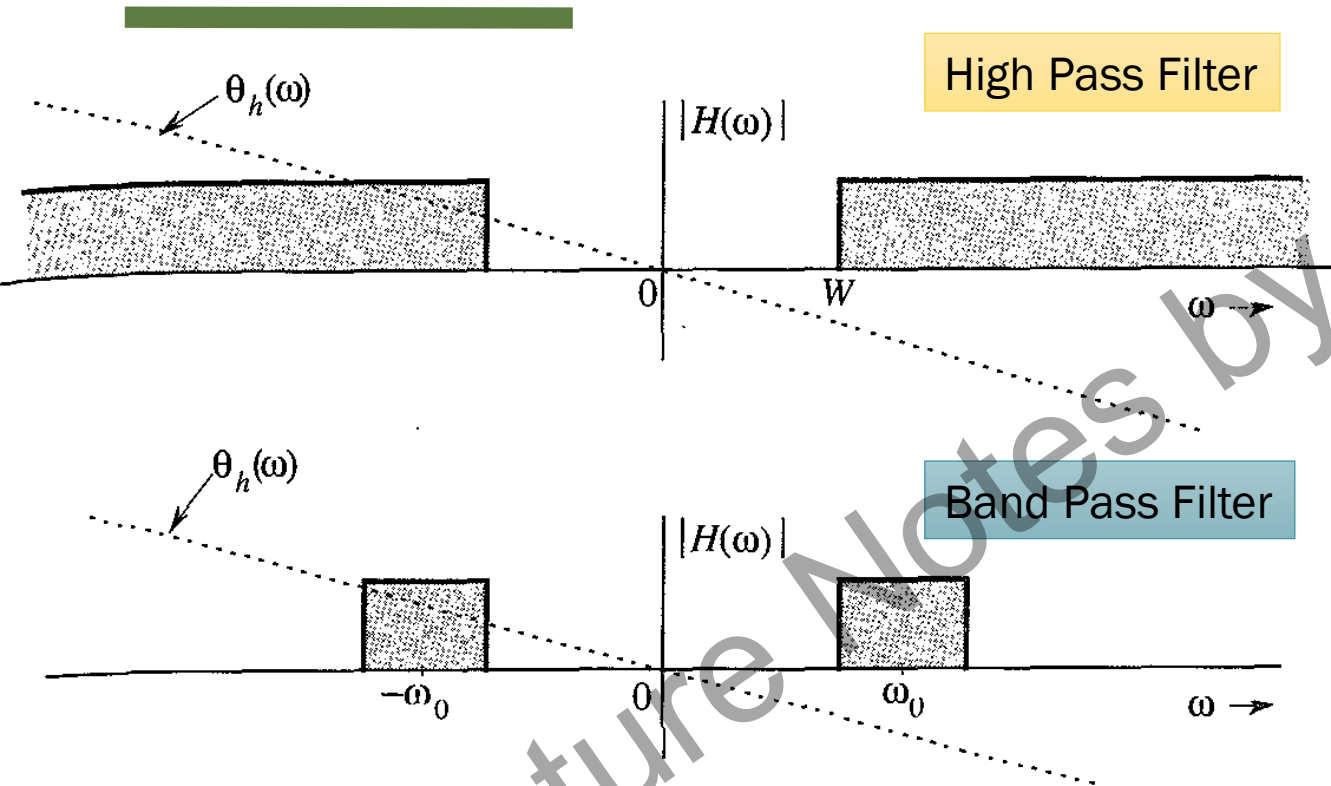
$$\begin{aligned}y(t) &= g(t - t_d) \\ H(\omega) &= |H(\omega)| e^{j\theta_h(\omega)} \\ H(\omega) &= \text{rect}\left(\frac{\omega}{2W}\right) e^{-j\omega t_d}\end{aligned}$$

Inverse Fourier Transform,

$$h(t) = \frac{W}{\pi} \text{sinc}[W(t - t_d)]$$

# Ideal and Practical Filters

Ideal filters **allow** distortionless transmission of a certain band of frequencies and **suppress** all the remaining frequencies as shown.

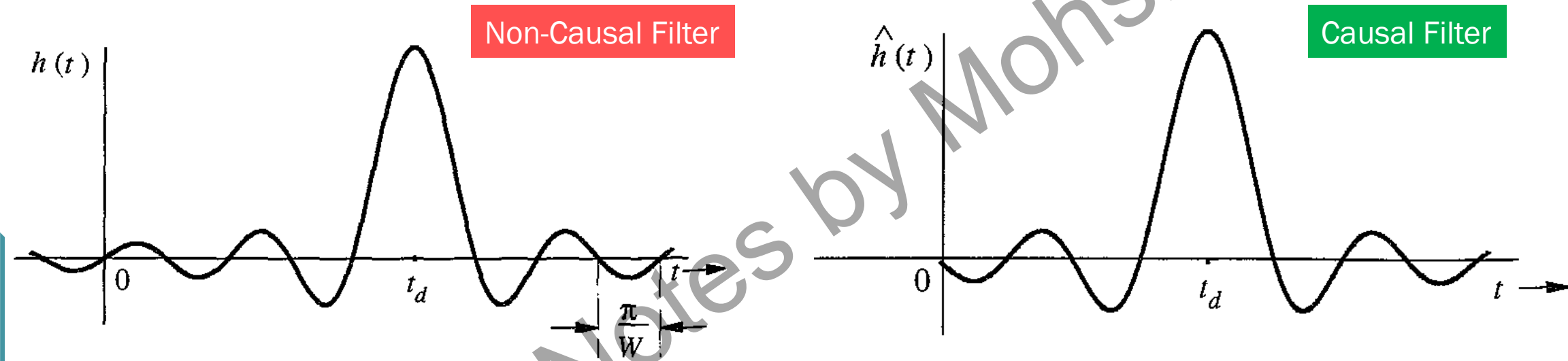


Inverse Fourier Transform,

$$h(t) = \frac{W}{\pi} \text{sinc} [W(t - t_d)]$$

# Ideal and Practical Filters

Ideal filters **allow** distortionless transmission of a certain band of frequencies and **suppress** all the remaining frequencies as shown.



Physically realizable filter

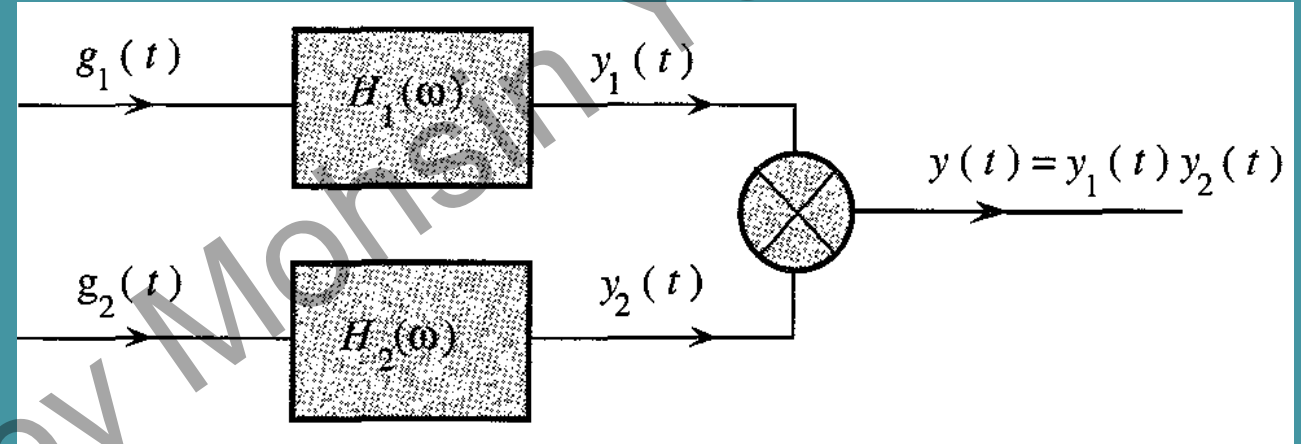
$$h(t) = 0 \quad \text{for } t < 0$$

$$\hat{h}(t) = h(t)u(t)$$

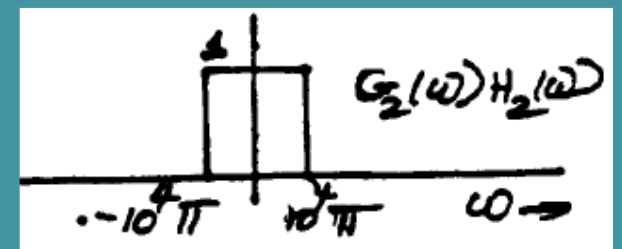
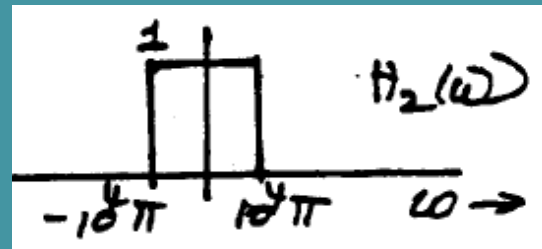
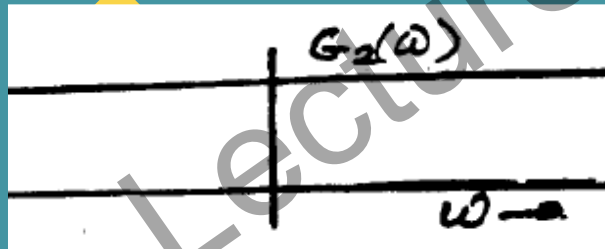
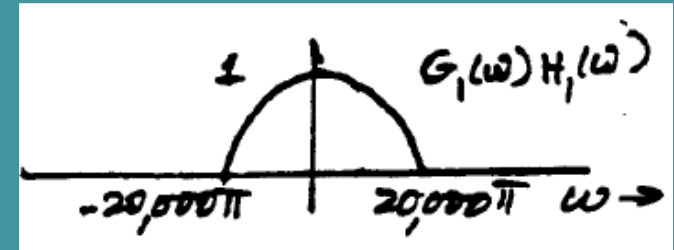
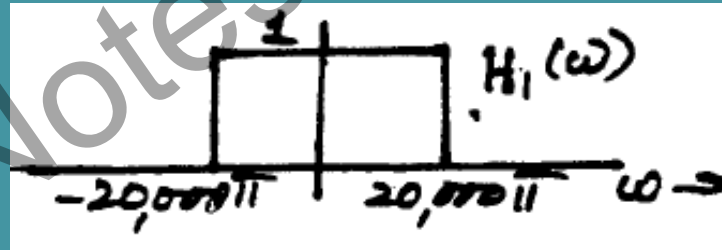
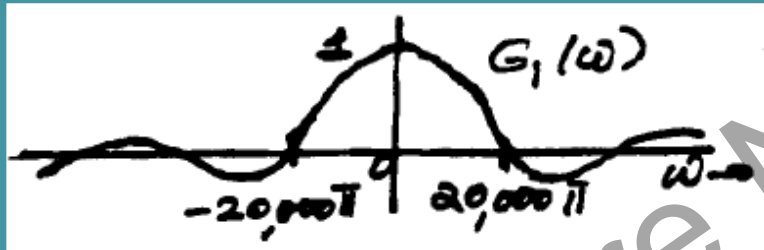
$$\int_{-\infty}^{\infty} \frac{|\ln |H(\omega)| |}{1 + \omega^2} d\omega < \infty$$

Signals  $g_1(t) = 10^4 \text{rect}(10^4 t)$  and  $g_2(t) = \delta(t)$  are applied at the inputs of the ideal low-pass filters  $H_1(\omega) = \text{rect}(\omega/40,000\pi)$  and  $H_2(\omega) = \text{rect}(\omega/20,000\pi)$  (Fig. P3.4-1). The outputs  $y_1(t)$  and  $y_2(t)$  of these filters are multiplied to obtain the signal  $y(t) = y_1(t)y_2(t)$ .

- Sketch  $G_1(\omega)$  and  $G_2(\omega)$ .
- Sketch  $H_1(\omega)$  and  $H_2(\omega)$ .
- Sketch  $Y_1(\omega)$  and  $Y_2(\omega)$ .
- Find the bandwidths of  $y_1(t)$ ,  $y_2(t)$ , and  $y(t)$ .



$$\text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$



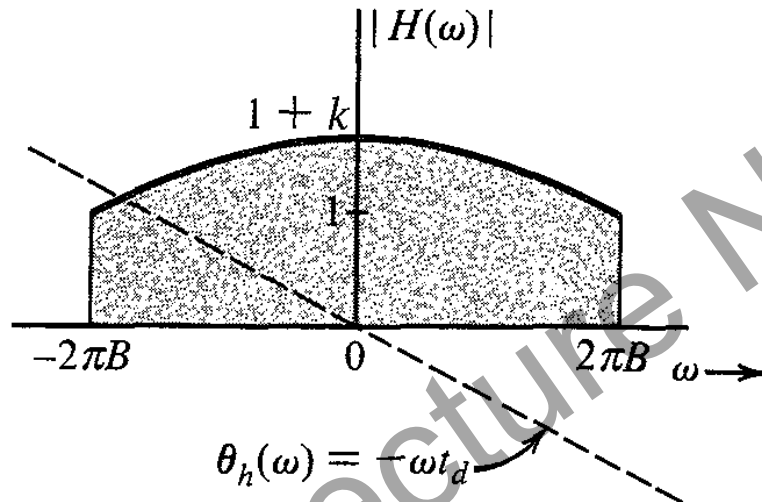
# Signal Distortion Over a Communication Channel

## Linear Distortion (Pulse Spreading)

**EXAMPLE 3.17** A low-pass filter (Fig. 3.33a) transfer function  $H(\omega)$  is given by

$$H(\omega) = \begin{cases} (1 + k \cos T\omega)e^{-j\omega t_d} & |\omega| < 2\pi B \\ 0 & |\omega| > 2\pi B \end{cases} \quad (3.60)$$

A pulse  $g(t)$  band-limited to  $B$  Hz (Fig. 3.33b) is applied at the input of this filter. Find the output  $y(t)$ .

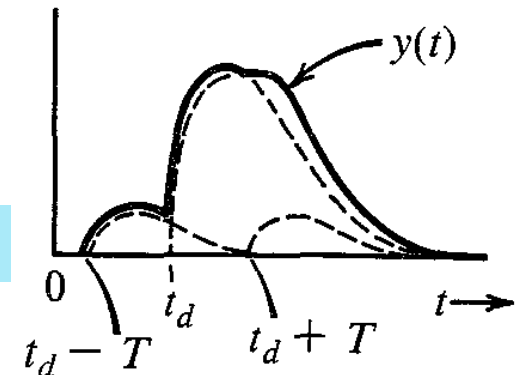
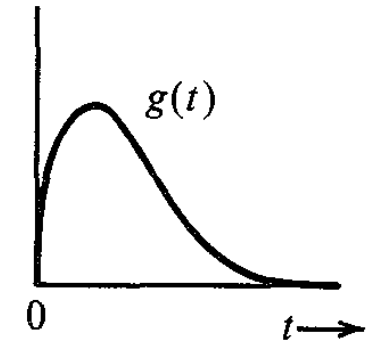


$$\begin{aligned} Y(\omega) &= G(\omega)H(\omega) \\ &= G(\omega)(1 + k \cos T\omega)e^{-j\omega t_d} \\ &= G(\omega)e^{-j\omega t_d} + k[G(\omega) \cos T\omega]e^{-j\omega t_d} \end{aligned}$$

Using the time-shifting property and Eqs. (3.30a) and (3.32), we have

$$y(t) = g(t - t_d) + \frac{k}{2}[g(t - t_d - T) + g(t - t_d + T)]$$

- The output is actually  $g(t) + \left(\frac{k}{2}\right)[g(t - T) + g(t + T)]$  delayed by  $t_d$  seconds.
- Ideal amplitude but nonideal phase response of  $H(\omega)$  has a similar effect.



# Signal Distortion Over a Communication Channel

## Distortion Caused by Channel Nonlinearities (Spectral Dispersion)

**EXAMPLE 3.18** The input  $x(t)$  and the output  $y(t)$  of a certain nonlinear channel are related as

$$y(t) = x(t) + 0.001x^2(t)$$

Find the output signal  $y(t)$  and its spectrum  $Y(\omega)$  if the input signal is  $x(t) = (1000/\pi) \text{sinc}(1000t)$ . Verify that the bandwidth of the output signal is twice that of the input signal. This is the result of signal squaring. Can the signal  $x(t)$  be recovered (without distortion) from the output  $y(t)$ ?

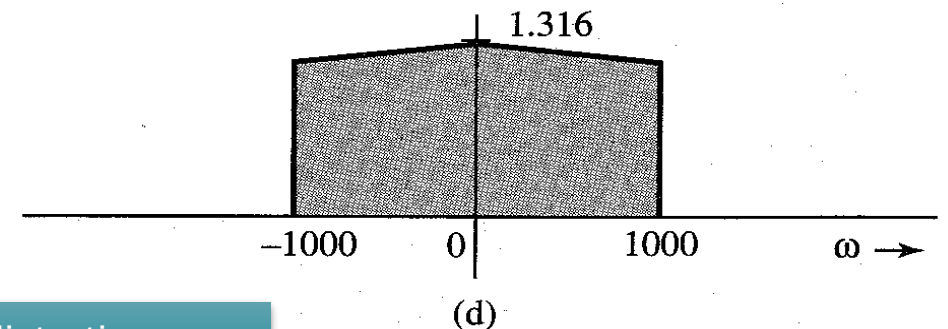
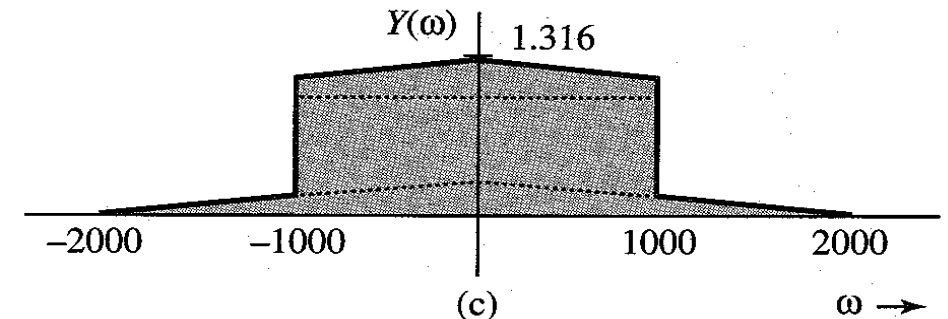
$$Y(\omega) = \text{rect}\left(\frac{\omega}{2000}\right) + 0.316 \Delta\left(\frac{\omega}{4000}\right)$$

The desired signal and the distorted signal spectra overlap, and it is impossible to recover  $x(t)$  from  $y(t)$  without some distortion.

Low Pass Filter (LPF)

FDM: Interference will occur along with spectral distortion.

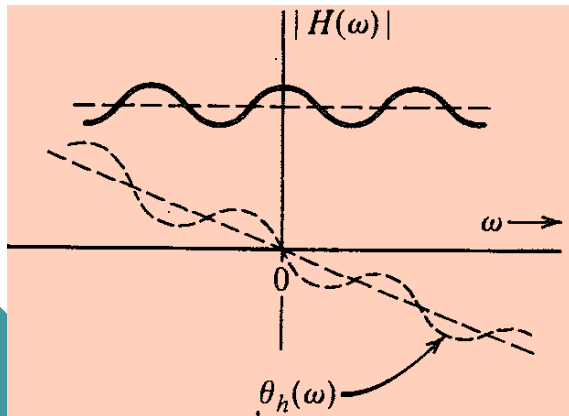
TDM: No Interference for two interleaved pulse trains along with pulse distortion.



# Signal Distortion Over a Communication Channel

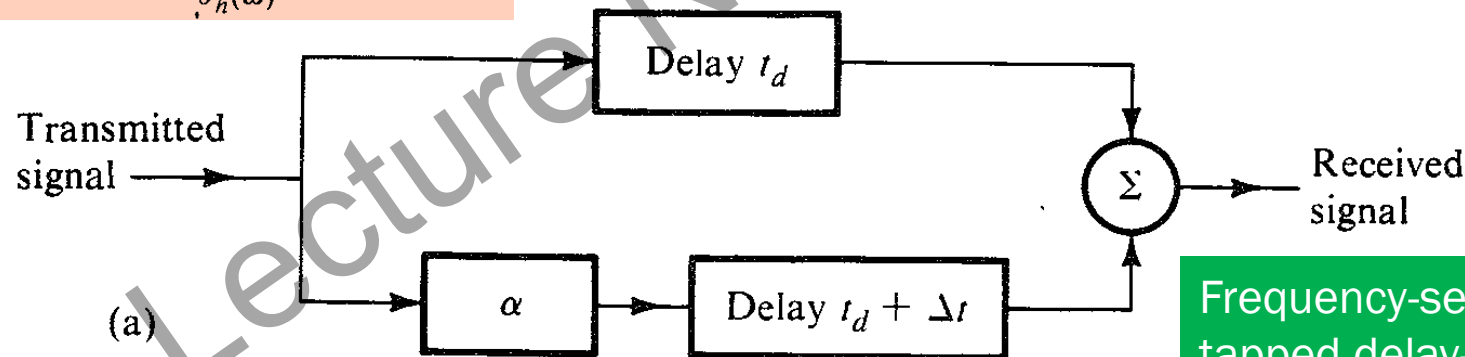
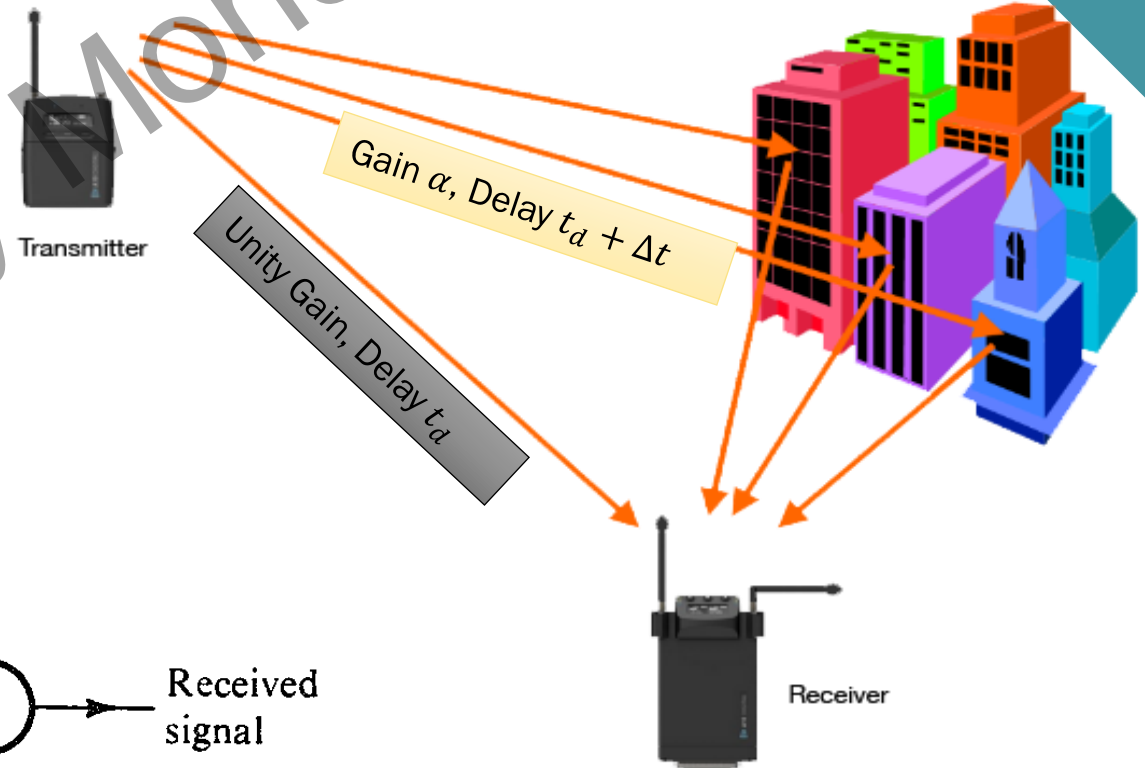
## Distortion Caused by Multipath Effects (Frequency-Selective Fading)

$$\begin{aligned} H(\omega) &= e^{-j\omega t_d} + \alpha e^{-j\omega(t_d + \Delta t)} \\ &= e^{-j\omega t_d} (1 + \alpha e^{-j\omega \Delta t}) \\ &= e^{-j\omega t_d} (1 + \alpha \cos \omega \Delta t - j\alpha \sin \omega \Delta t) \\ &= \sqrt{1 + \alpha^2 + 2\alpha \cos \omega \Delta t} e^{-j \left[ \omega t_d + \tan^{-1} \frac{\alpha \sin \omega \Delta t}{1 + \alpha \cos \omega \Delta t} \right]} \end{aligned}$$



Periodic in  $\omega$  with a period of  $2\pi/\Delta t$  causes pulse dispersion.

Delay in time-domain corresponds to multiplication by an exponential in frequency domain



Frequency-selective fading can be corrected by tapped delay-line equalizer